Introduction to Probability and Statistics

(EE 354 / CE 361)

Counting: Part 1

Today's Agenda

- The Counting Principle
- Permutations
- Combinations
- Partitions

Rules of Counting



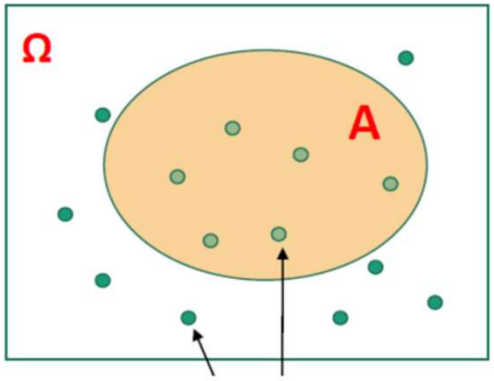
Counting: Why do we do it?

- The calculation of probabilities often involves counting the number of outcomes in various events
- For example (Discrete Uniform Law):

When the sample space Ω has a finite number of equally likely outcomes, so that the discrete uniform probability law applies. Then, the probability of any event A is given by

$$\mathbf{P}(A) = \frac{\text{number of elements of } A}{\text{number of elements of } \Omega},$$

and involves counting the elements of A and of Ω.



Count the number of ways two independent events can be performed

If event A can occur in n1 ways and event B can occur in n2 ways, then the number of ways these events can occur together = n1*n2

Example : Rules of Counting

- Suppose that you want to dress up by choosing from 10 formal shirts (tuxedo shirts), 15 dress pants, and 7 neckties.

  You can dress up in 10*15*7=1050 ways ☺

The Counting Principle

The Counting Principle

Consider a process that consists of r stages. Suppose that:

(a) There are n_1 possible results at the first stage.

(b) For every possible result at the first stage, there are n_2 possible results at the second stage.

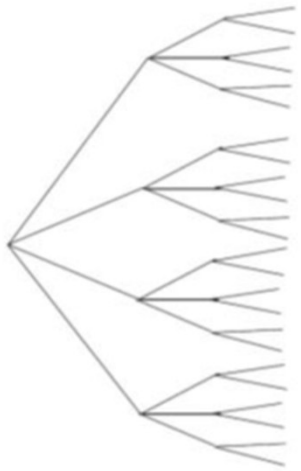
(c) More generally, for any sequence of possible results at the first $i-1$ stages, there are n_i possible results at the ith stage. Then, the total number of possible results of the r-stage process is

$$n_1 n_2 \cdots n_r.$$

For example:

- 4 shirts
- 3 ties
- 2 jackets

- What are the number of possible attires?
  - r stages
  - n_i choices at stage i



The Counting Principle: Example

Find the probability that six rolls of 6-sided die all give different numbers. Assume all outcomes are equally likely.

$$\mathbf{P}(A) = \frac{\#\ in\ A}{\#\ possible\ outcomes}$$

Factorial Operator

- The Symbol n! is read as factorial of n and is computed as follows
- n!= n(n-1)(n-2)(n-3).....(3)(2)(1)
- 6!=(6)(5)(4)(3)(2)(1)=720

Example. The simplest protein molecule in biology is called *vasopressin* and is composed of 8 amino acids that are chemically bound together in a particular order. The order in which these amino acids occur is of **vital importance** to the proper functioning of vasopressin. If these 8 amino acids were placed in a hat and drawn out randomly one by one, how many different arrangements of these 8 amino acids are possible?

Solution: Let A,B,C,D,E,F,G,H symbolize the 8 amino acids. They must fill 8 slots: ___ ___ ___ ___ ___ ___ ___ ___ . There are 8 choices for the first position, leaving 7 choices for the second slot, 6 choices for the third slot and so on. The number of different orderings is

8(7)(6)(5)(4)(3)(2)(1)=8! =40,320.

Combination Vs. Permutation

Definition	Formula	Example
A *combination* is a grouping of outcomes in which the order does not matter.	The number of *combinations* of *n* things chosen *r* at a time is found using $_nC_r = \frac{n!}{r!(n-r)!}$.	How many pairs can be made from a group of 6 people? (Jin and Tom are the same pair as Tom and Jin) $_6C_2 = \frac{6!}{2!(6-2)!} = \frac{720}{2(24)} = 15$
Definition	**Formula**	**Example**
A *permutation* is an arrangement of outcomes in which the order does matter.	The number of *permutations* of *n* things chosen *r* at a time is found using $_nP_r = \frac{n!}{(n-r)!}$.	6 people are in a contest. How many ways can 1st and 2nd place be awarded? (Jin first and Tom second is different than Tom first and Jin second) $_6P_2 = \frac{6!}{(6-2)!} = \frac{720}{24} = 30$

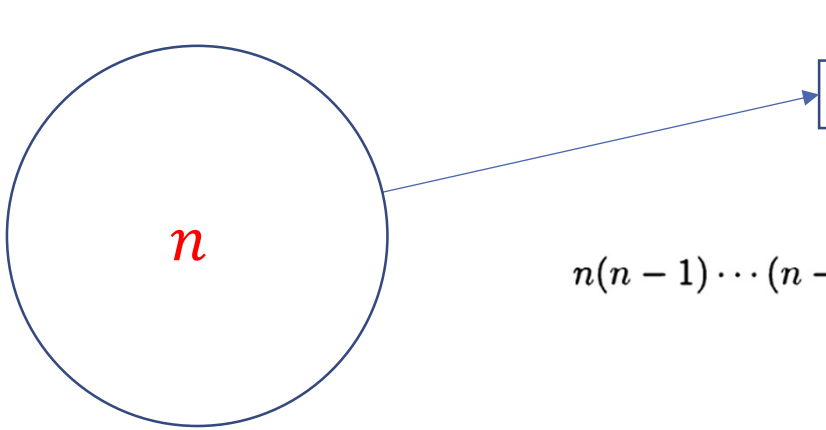
Combinations

Permutations

k-Permutations

We start with n distinct objects, and let k be some positive integer, with $k \leq n$. We wish to count the number of different ways that we can pick k out of these n objects and arrange them in a sequence, i.e., the number of distinct k-object sequences.

- Process
  - We can choose any of the n objects to be the first one.
  - Having chosen the first one, there are only $n-1$ possible choices for the second.
  - Having chosen the first two, there remain only $n-2$ available objects for the third stage.
  - When we are ready to select the last (kth object), we have already chosen $k-1$ objects, which leaves us with $n-(k-1)$ choices for the last one. Consider the selection of k objects out of a collection of n objects.





$$n(n-1)\cdots(n-k+1) = \frac{n(n-1)\cdots(n-k+1)(n-k)\cdots 2\cdot 1}{(n-k)\cdots 2\cdot 1}$$
$$= \frac{n!}{(n-k)!}.$$

k-Permutations: Example

Example 1.28. Let us count the number of words that consist of four distinct letters. This is the problem of counting the number of 4-permutations of the 26 letters in the alphabet. The desired number is

$$\frac{n!}{(n-k)!} = \frac{26!}{22!} = 26 \cdot 25 \cdot 24 \cdot 23 = 358,800.$$

CIRCULAR PERMUTATIONS

a. Number of circular arrangements (permutations) of n different things = $(n-1)!$

b. Number of circular arrangements (permutations) of n different things when clockwise and anticlockwise arrangement are not different i.e. when same observation can be made from both

sides = $\frac{1}{2}(n-1)!$.

c. Number of circular permutations of n different things, taken r at a time, when clockwise, and anticlockwise orders are taken as different, is = $\frac{{}^{n}P_{r}}{r}$.

d. Number of circular permutations of n different thins, taken r at a time, when clockwise an anticlockwise orders are not different, is = $\frac{{}^{n}P_{r}}{2r}$.

Permutation Example 1

Finding Permutations of n objects taken r at a time

You have 12 songs downloaded on your computer and want to put four of them on a CD. How many different ways can you do this? Leave the commas out of your answer.

___ ways

$${}_nP_r = \frac{n!}{(n-r)!} \qquad n = 12 \qquad r = 4$$

$${}_{12}P_4 = \frac{12!}{(12-4)!} = 11880$$

Permutation Example 2

Example 2

A permutation lock will open when the right choice of three numbers (from 1 to 30, inclusive) is selected. How many different lock permutations are possible assuming no number is repeated?

$$_{30}P_3 = \frac{\mathbf{30!}}{\mathbf{(30-3)!}} = \frac{30!}{27!} = \mathbf{30*29*28 = 24360}$$

PERMUTATIONS

Order Matters
Repetition Allowed

$$Possibilities = n^r$$

Order Matters
Repetition Not Allowed

$$Possibilities = \frac{n!}{(n-r)!}$$

COMBINATIONS

Order Doesn't Matter
Repetition Allowed

$$Possibilities = \frac{n!}{r!\,(n-r)!}$$

Order Doesn't Matter
Repetition Not Allowed

$$Possibilities = \frac{(n+r-1)!}{r!\,(n-1)!}$$

Order Matters, Repetition allowed

- Suppose that you want to analyze in how many ways can a car registration number (having 4 digits) be issued.
- We know that there are total 10 digits (0, 1,2,…..9)
- Note that repetition is possible (for instance a car registration number may be 0112). Order matters because 1122 is different from 2211.
- So, total

$$(10)^4 = 10 * 10 * 10 * 10$$
$$= 10{,}000\ ways$$

Order Matters, Repetition allowed

- Suppose that you want to analyze in how many ways can a car registration number (having 4 digits) be issued. Suppose that a digit can only be used only once.
- We know that there are total 10 digits (0, 1,2,…..9)
- Note that repetition is not possible. Order matters because 1234 is different from 4123 or 2341.
- So, total ways are

$$^{10}P_4 = \frac{10!}{(10-4)!} = \frac{10.9.8.7.6.5.4.3.2.1}{6.5.4.3.2.1} = 10.9.8.7$$
$$= 5040\ ways$$

Two problems illustrating combinations and permutations.

Consider the following two problems:

1) Consider the set { p , e , n} How many two-letter "words" (including nonsense words) can be formed from the members of this set?

We will list all possibilities: pe, pn, en, ep, np, ne , a total of 6.

2) Now consider the set consisting of three males: {Paul, Ed, Nick} For simplicity, we will denote the set { p, e, n} How many two-man crews can be selected from this set?

3) Answer: pe (Paul, Ed), pn (Paul, Nick) and en (Ed, Nick) and that is all!

Difference between permutations and combinations

- The difference between the two problems is this:
- Both problems involved counting the numbers of arrangements of the same set {p , e , n}, taken 2 elements at a time, without allowing repetition. However, in the first problem, the **order of the arrangements mattered** since pe and ep are two different "words". In the second problem, the **order did not matter** since pe and ep represented the same two-man crew. So we counted this only once.
- The first example was concerned with counting the number of **permutations** of 3 objects taken 2 at a time.

- The second example was concerned with the number of **combinations** of 3 objects taken 2 at a time

Combinations

- Problem of counting the number of k-element subsets of a given n-element set.

- How is this different from k-permutations?

- Example:
  - There are n people and we are interested in forming a committee of k. How many different committees are possible?

$$\binom{n}{k}$$

Combinations

- Problem of counting the number of k-element subsets of a given n-element set.

$$\binom{n}{k}$$

Notice that forming a combination is different than forming a k-permutation, because **in a combination there is no ordering of the selected elements**. For example, whereas the 2-permutations of the letters A, B, C, and D are

AB. BA, AC, CA, AD, DA, BC. CB, BD. DB, CD, DC,

the combinations of two out of these four letters are

AB, AC, AD, BC, BD. CD.

In the preceding example, the combinations are obtained from the permutations by grouping together "duplicates"; for example, AB and BA are not viewed as distinct, and are both associated with the combination AB. This reasoning can be generalized: each combination is associated with $k!$ "duplicate" k-permutations

$$\binom{n}{k} k! = \frac{n!}{(n-k)!}$$

$$\boxed{\binom{n}{k} = \frac{n!}{k!\,(n-k)!}}$$

Combinations: Example

Example 1.30. The number of combinations of two out of the four letters A, B. C, and D is found by letting $n = 4$ and $k = 2$. It is

$$\binom{4}{2} = \frac{4!}{2!\,2!} = 6,$$

Combinations: Interesting Cases

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

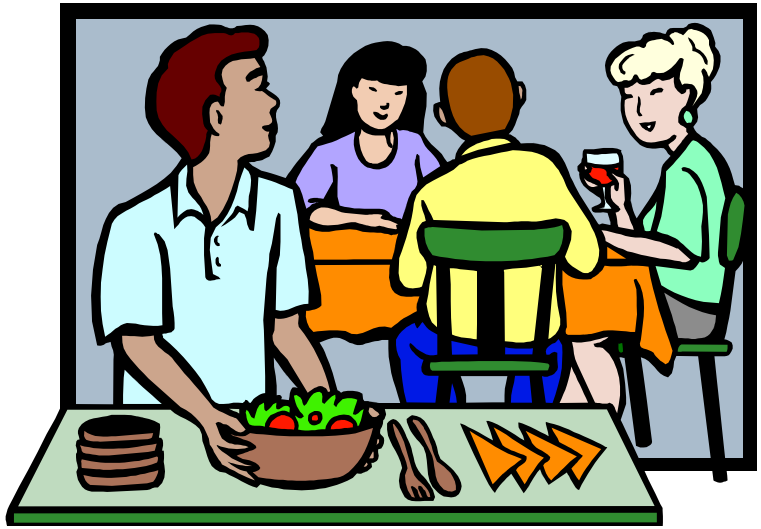
$$\binom{n}{n} = \frac{n!}{n!\,(n-n)!} = \frac{n!}{n!\,0!} = 1 \qquad 0! = 1$$

$$\binom{n}{0} = \frac{n!}{0!\,(n-0)!} = 1$$

$$\sum_{k=0}^{n} \binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n$$

Combinations or Permutations?

- 1. In how many ways can you choose 5 out of 10 friends to invite to a dinner party?
- Solution: Does the order of selection matter? If you choose friends in the order A,B,C,D,E or A,C,B,D,E the same set of 5 was chosen, so we conclude that the order of selection does not matter. We will use the formula for combinations since we are concerned with how many **subsets of size 5** we can select from a set of 10.
- C(10,5) =



$$\frac{P(10,5)}{5!} = \frac{10(9)(8)(7)(6)}{5(4)(3)(2)(1)} = \frac{10(9)(8)(7)}{(5)(4)} = 2(9)(2)(7) = 252$$

Example:

Six friends want to play enough games of chess to be sure every one plays everyone else. How many games will they have to play?

Solution:

There are 6 players to be taken 2 at a time.

Using the formula:

$$C(6,2) = \frac{P(6,2)}{2!} = \frac{6 \times 5}{2 \times 1} = 15$$

They will need to play 15 games.

In a lottery, each ticket has 5 one-digit numbers 0-9 on it.

a) You win if your ticket has the digits in any order. What are your changes of winning?

b) You would win only if your ticket has the digits in the required order. What are your chances of winning?

Solution:

There are 10 digits to be taken 5 at a time.

a) Using the formula:

$$C(10,5) = \frac{P(10,5)}{5!} = \frac{10 \times 9 \times 8 \times 7 \times 6}{5!} = 252$$

The chances of winning are 1 out of 252.

b) Since the order matters, we should use permutation instead of combination.

P(10, 5) = 10 x 9 x 8 x 7 x 6 = 30240

The chances of winning are 1 out of 30240.

Problem 1

How many total arrangements of the letters in MISSISSIPPI are there?

Since MISSISSIPPI has 11 letters, draw eleven lines and fill each in with the number of available letter choices, e.g. 11 options for the first, 10 for the second, and so on…

$$\underline{11} \cdot \underline{10} \cdot \underline{9} \cdot \underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1}$$

This is equal to 11! or 39,916,800 permutations.

Problem 1

How many total arrangements of the letters in MISSISSIPPI are there?

Consider this...

What happens if I switch the 3rd and 4th letters in MISSISSIPPI and leave all else the same? Are those different permutations?

Nope.

The 3rd and 4th letters are both S's so switching them changes nothing. The above calculation is an over count because of these repeat letters.

So how do we adjust for these extra repeated arrangements?

Problem 1

How many total arrangements of the letters in MISSISSIPPI are there?

$$\frac{11!}{2!4!4!} = 34{,}650$$

There we go! There are 34,650 permutations of the word MISSISSIPPI.

Problem 2

How many arrangements of the letters in MISSISSIPPI exist such that the 2 P's are separated?

Solution #2: No Adjacent P's

To solve this problem we have to get a little creative. We need to count the ways we can make permutations so that no P's are adjacent. Let's start simple and remove the P's altogether.

M I S S I S S I I

Start by removing the P's & calculating the permutations

This of course is just one arrangement of MISSISSII, how many ways could we rearrange those letters? Using the Multinomial Theorem there are **9!/(4!4!) or 630 permutations.**

Okay, now let's deal with the P's. I'm going to draw in blanks before and after every letter of MISSISSII. These will represent the possible places we can insert P's, and because each space is separated by some non-P letter, we know these are all safe possibilities.

_ M _ I _ S _ S _ I _ S _ S _ I _ I _

We have 10 blanks and we can choose any 2 blanks to insert P's at, so we need to calculate **10 choose 2 = 45 ways** (click here for recap on the "choose" combinations formula).

Alright, last step.

We calculated that there are 630 ways of rearranging the non-P letters and 45 ways of inserting P's, so to find the total number of desired permutations use the basic principle of counting, i.e. multiply the values together.

$$630 \cdot 45 = 28{,}350$$

Courtesy: https://medium.com/i-math/can-you-solve-the-mississippi-problem-6c0f3b02531

Practice Problem 1

"A certain class of 20 HU students are asked to undertake a class project comprising of two students. A geek HU programmer decides to write a Python program that lists all possible groups (N_1 = Number of possible groups) that can possibly be formed. She takes her program to a new level and based on some crude AI also predicts the group experience U_1 - based on student profiles and maps this on a scale of 3 (1-crappy, 2-ok, 3-excellent)"

What is N_1?

"Based on a very poor feedback with regards to the class project exercise, in the next offering the instructor now asks each group to select a group lead & a member within each group. Assuming that this class also has 20 HU students, the number of possible group curations, N_2, are now mapped (using the same crappy AI) to predict the user experience U_2"

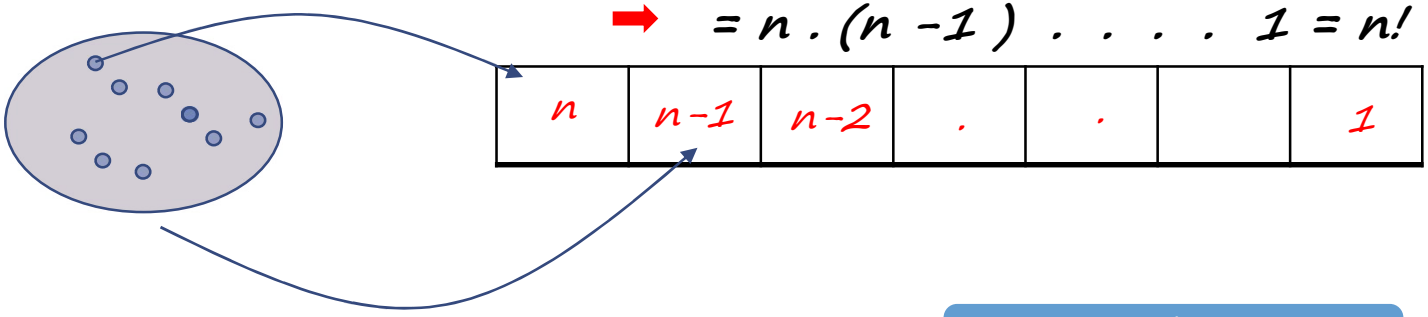
What is N_2?

After group activity – Discuss U_1 versus U_2

Basic Counting Principle (ctd)

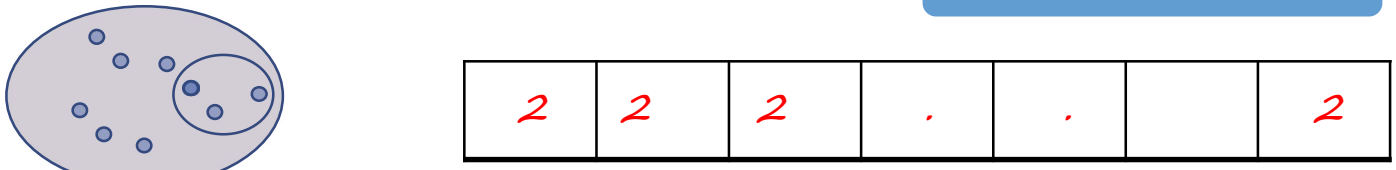
Example:

- **Number of license plates with 2 letters followed by 3 digits:** ➡ $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10$
  - What if repetition is prohibited?
    ➡ $26 \cdot 25 \cdot 10 \cdot 9 \cdot 8$
- **Permutations**: *Number of ways of ordering n elements*
  ➡ $= n \cdot (n-1) \cdot \ldots \cdot 1 = n!$



- **Number of distinct subsets of {1, … , n}:** $= 2^n$ Remember this!

?



Example

- Find the probability that:
  six rolls of a (six sided) die all give different numbers.
  (Assume all outcomes equally likely.)

A

➡ *Typical outcome:* $P(2, 3, 4, 5, 6, 2) = \frac{1}{6^6}$

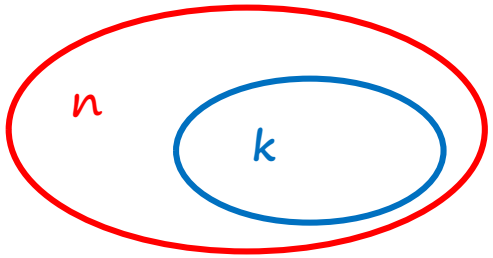
➡ *Typical element of A:* $(2, 3, 4, 1, 6, 5) = 6!$

➡ $$P(A) = \frac{\#\ in\ A}{\#\ possible\ outcomes} = \frac{6!}{6^6}$$

Combinations

Definition: $\binom{n}{k}$ Number of *k*-element subsets of a given *n*-element set

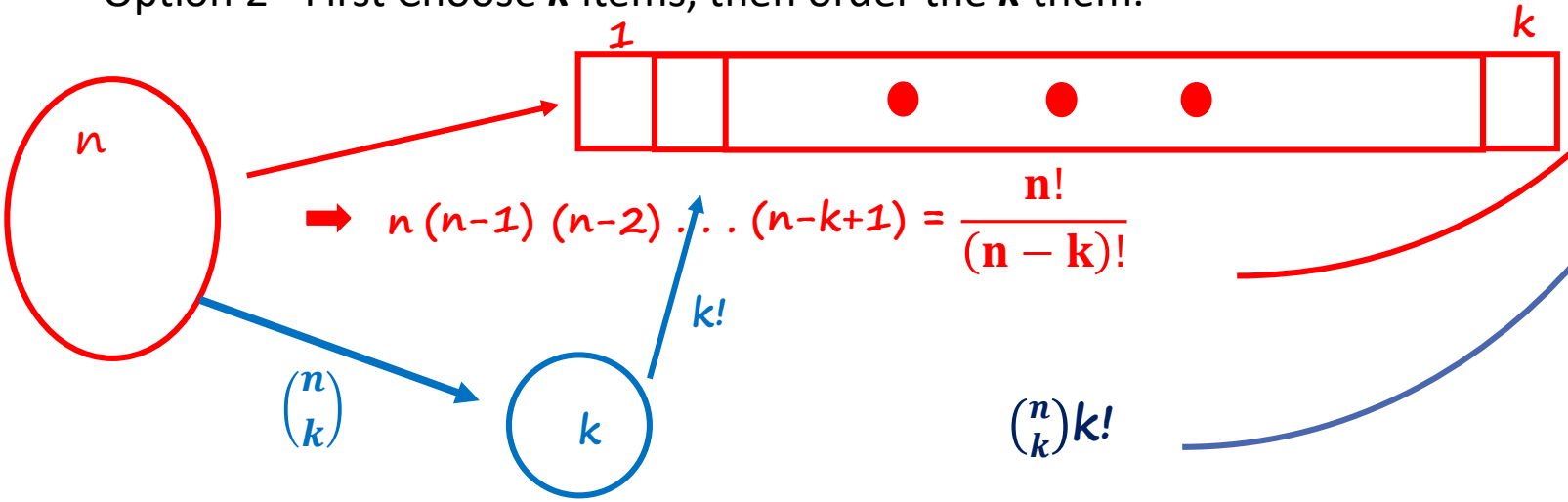
$$= \binom{n}{k} = \frac{n!}{k!(n-k)!}$$



➡ n = 0, 1, 2, …

➡ k = 0, 1, 2, … , m

- Two ways of constructing an **ordered** sequence of *k* **distinct** items:
  - Option 1 - Choose the ***k*** items one at a time.
  - Option 2 - First Choose ***k*** items, then order the ***k*** them.

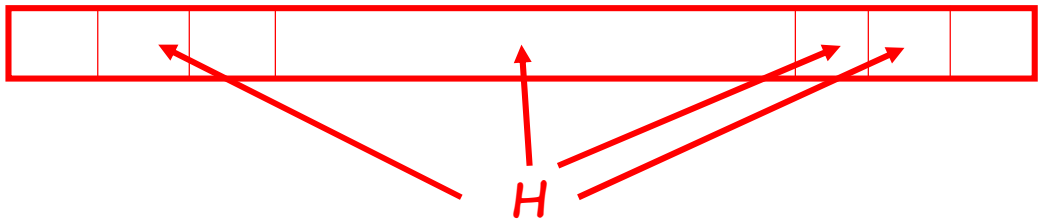


Binomial coefficient $\binom{n}{k}$ → Binomial probabilities

$$\mathbf{P}\ (k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- n >= 1 independent coin tosses; $\mathbf{P}\ (H) = p$
- $\mathbf{P}\ (H\ T\ T\ H\ H\ H) = p\ (1-p)\ (1-p)\ p\ p\ p = p^4\ (1-p)^2$ (n=6)
- $\mathbf{P}\ (\textit{particular sequence}) = p^{\#heads}\ (1-p)^{\#tails}$
- $\mathbf{P}\ (\textit{particular k-head sequence}) = p^k\ (1-p)^{n-k}$

➡ $\mathbf{P}\ (k \text{ heads}) = p^k\ (1-p)^{n-k}$ * (# of k-head sequence)



$\binom{n}{k}$

A coin tossing problem

Assumptions:

- **independence**
- $\mathbf{P}(H) = p$

$$\mathbf{P}(k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?
  - event *A* : the first 2 tosses were heads.
  - event *B* : 3 out of 10 tosses were heads.

- First solution:

➡ $$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} = \frac{P(H_1\, H_2 \text{ and one } H \text{ in (between) tosses } 3, 4, \ldots, 10)}{P(B)}$$

➡ $$= \frac{p^2 . \binom{8}{1} p^1 . (1-p)^7}{\binom{10}{3} p^3 (1-p)^7} = \frac{\binom{8}{1}}{\binom{10}{3}} = \frac{8}{\binom{10}{3}}$$

A coin tossing problem

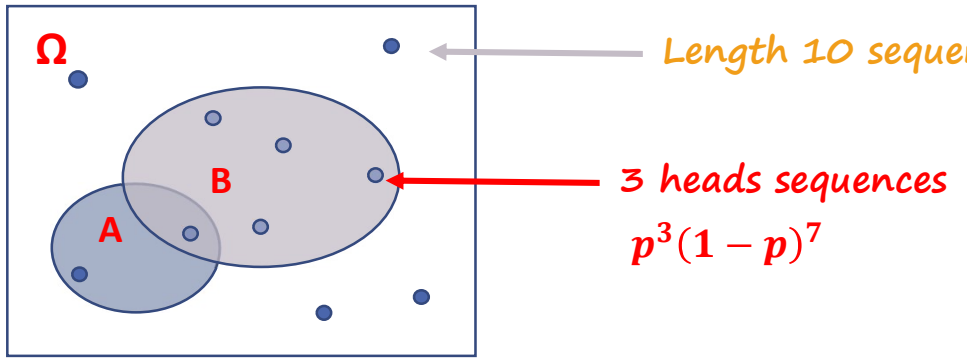
Assumptions:

- **independence**
- **P** (*H*) = p

$$\mathbf{P}\ (k \text{ heads}) = \binom{n}{k} p^k (1-p)^{n-k}$$

- Given that there were 3 heads in 10 tosses, what is the probability that the first two tosses were heads?
  - event *A* : the first 2 tosses were heads.
  - event *B* : 3 out of 10 tosses were heads.

- Second solution: Conditional probability law (on *B*) is uniform



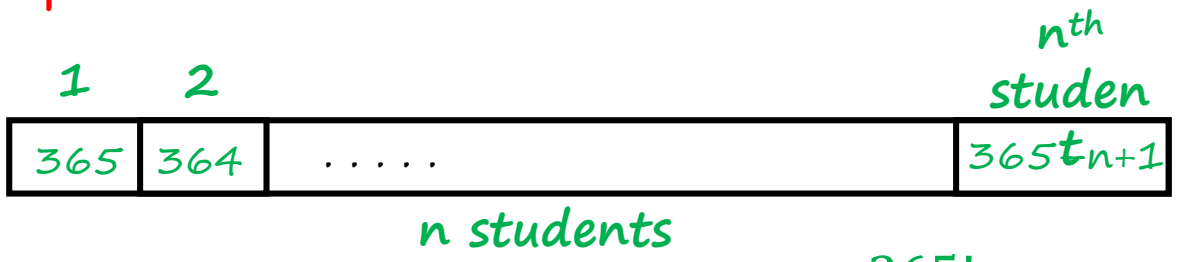
$$\frac{\#\ in\ (A \cap B)}{\#\ in\ B} = \frac{8}{\binom{10}{3}}$$

Example: The Birthday Problem

- Given there are n Students in a class, find the P (*no 2 birthdays coincide*)
  - $1 < n < 365$

Ω = all possible sequences

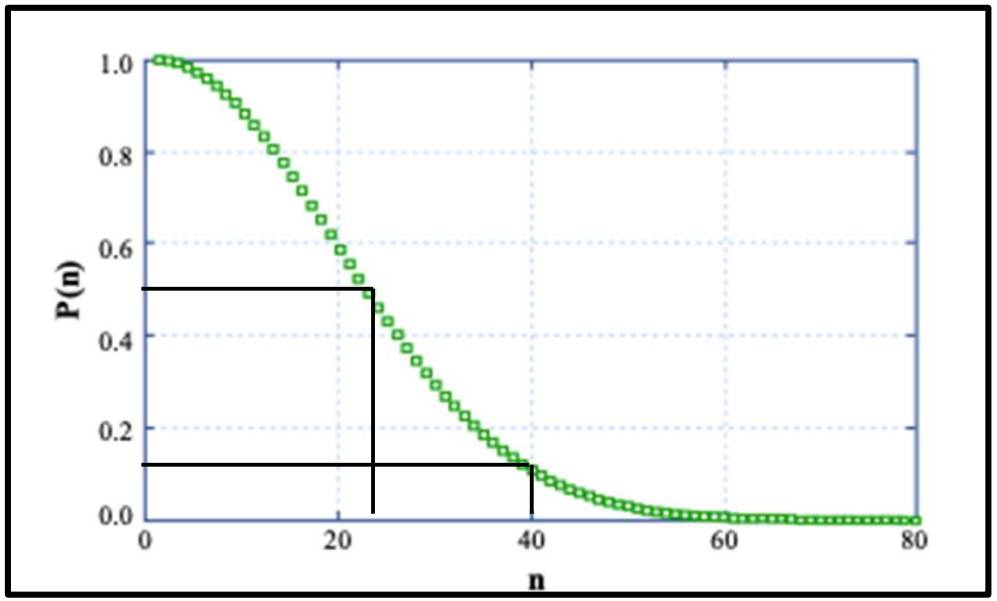
$|\Omega| = (365)^n$



$S = 365 * 364 * .... * (365-n+1) = \frac{365!}{(365-n)!}$

P (*no 2 birthdays coincide*) $= \frac{365!}{(365-n)!}\frac{1}{(365)^n}$

Example: The Birthday Problem (ctd)



$$\mathbf{P}(\textit{ no 2 birthdays coincide }) = \frac{365!}{(365-n)!}\frac{1}{(365)^n}$$

Partitions

- $n \geq 1$ distinct items; $r \geq 1$ persons

  Give n_i items to person i

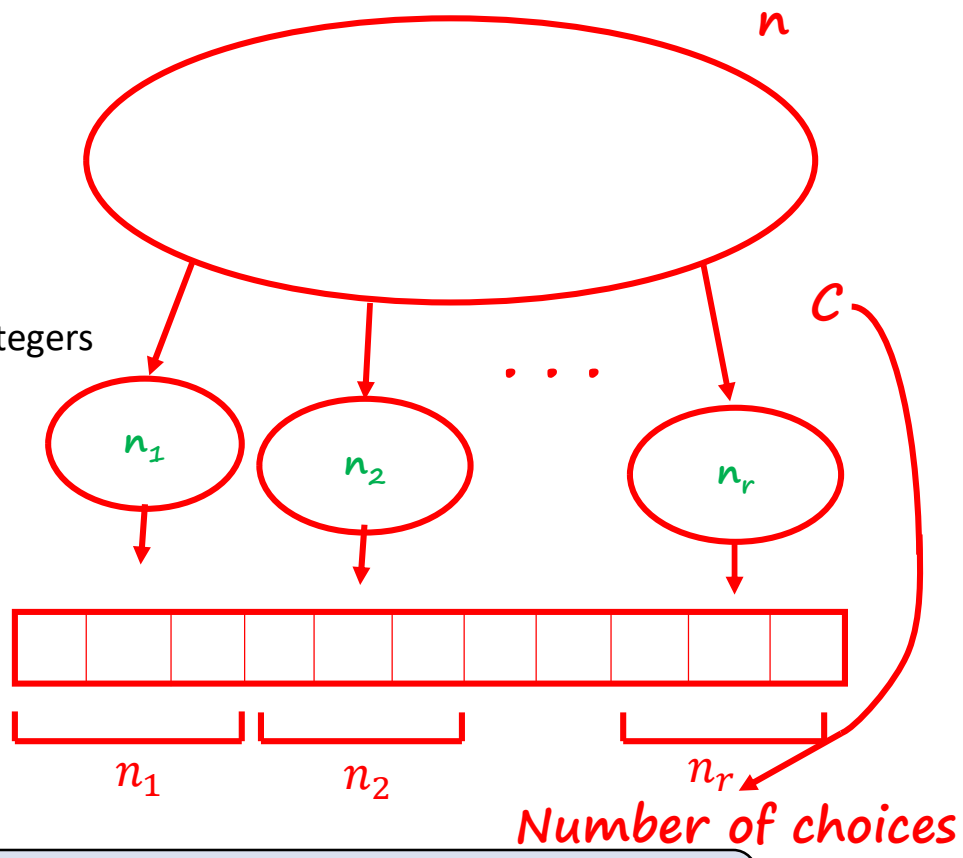
  - Here $n_1, .., n_r$ are given nonnegative integers
  - With $n_1 + \cdots + n_r = n$ are given nonnegative integers

- Ordering n items: $n!$

  - Deal n_i to each person i, and then order

$$C.\, n_1!\, n_2!\, \ldots n_r! = n!$$

Solving for C we get:



$$\text{Number of Choices, } C = \frac{n!}{n_1!\, n_2!\, \ldots n_r!} \quad \text{(multinomial coefficient)}$$

Partitions

We are given an n-element set and nonnegative integers $n_1, n_2, \ldots, n_r$, whose sum is equal to n. We consider partitions of the set into r disjoint subsets. with the ith subset containing exactly n_i elements. Let us count in how many ways this can be done.

We form the subsets one at a time. We have $\binom{n}{n_1}$ ways of forming the first subset. Having formed the first subset, we are left with $n - n_1$ elements. We need to choose n_2 of them in order to form the second subset, and we have $\binom{n-n_1}{n_2}$ choices, etc. Using the Counting Principle for this r-stage process, the total number of choices is

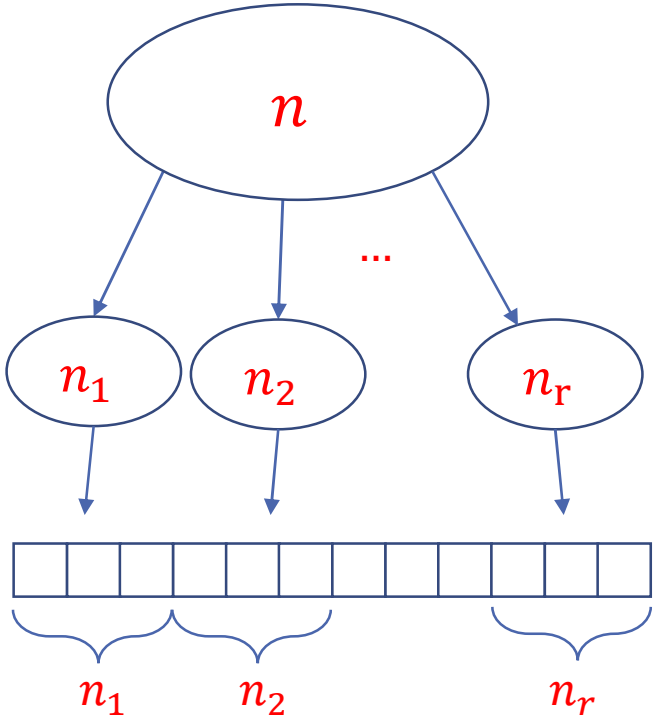
$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r},$$

which is equal to

$$\frac{n!}{n_1!\,(n-n_1)!}\cdot\frac{(n-n_1)!}{n_2!\,(n-n_1-n_2)!}\cdots\frac{(n-n_1-\cdots-n_{r-1})!}{(n-n_1-\cdots-n_{r-1}-n_r)!\,n_r!}.$$

We note that several terms cancel and we are left with

$$\frac{n!}{n_1!\,n_2!\cdots n_r!}.$$



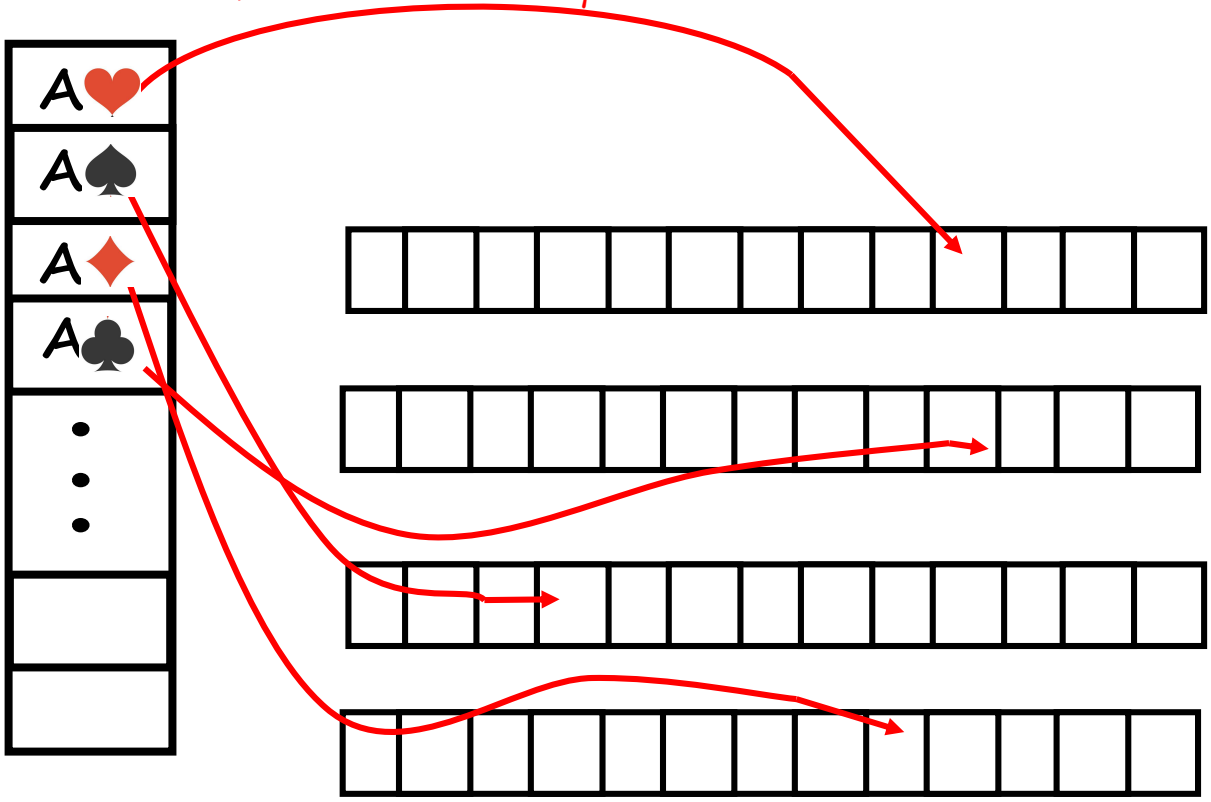
Example: **52-card deck, dealt (fairly) to four players.**
Find P(each player gets an ace)

- Total possible Number of outcomes: $\frac{52!}{13!\,13!\,13!\,13!}$
- Constructing an outcome with one ace to each person: (Distribute the aces) $\frac{4!}{1!\,1!\,1!\,1!} = 4 * 3 * 2 * 1$
- Distribute the remaining 48 cards: $\frac{48!}{12!\,12!\,12!\,12!}$
- Answer: P(each player gets an ace) = $\frac{\frac{4!}{1!\,1!\,1!\,1!} * \frac{48!}{12!\,12!\,12!\,12!}}{\frac{52!}{13!\,13!\,13!\,13!}}$

Example: **52-card deck, dealt (fairly) to four players.**
Find P(each player gets an ace)

A smart solution

Stack the deck, with 4 aces on top



$$\frac{A\heartsuit}{52} * \frac{A\spadesuit}{39} * \frac{A\diamondsuit}{26} * \frac{A\clubsuit}{13}$$

$$\frac{52}{52} * \frac{39}{51} * \frac{26}{50} * \frac{13}{49}$$

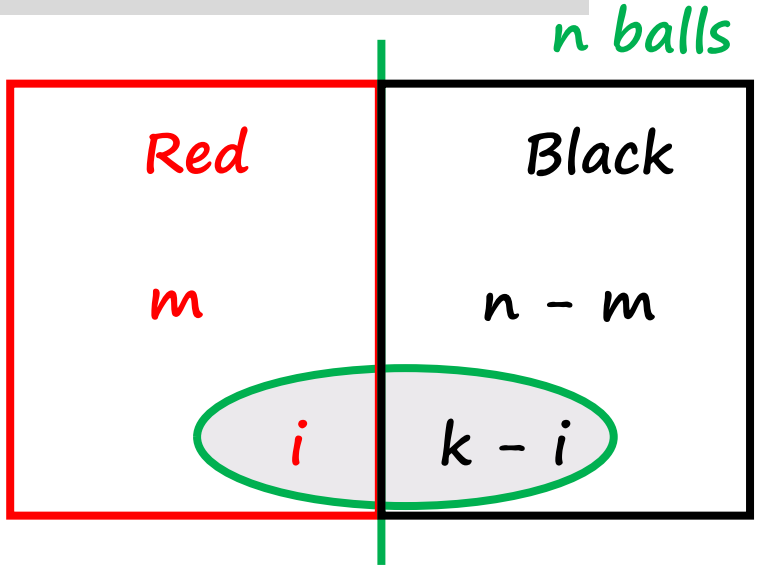
$$= 0.105$$

Counting Example

We have a total of n balls, out of which m are red balls. The remaining (n-m) balls are black balls

➡ We have to pick k balls such that i balls are red

Find the Probability,
P (i red balls) = P_r



Counting Example

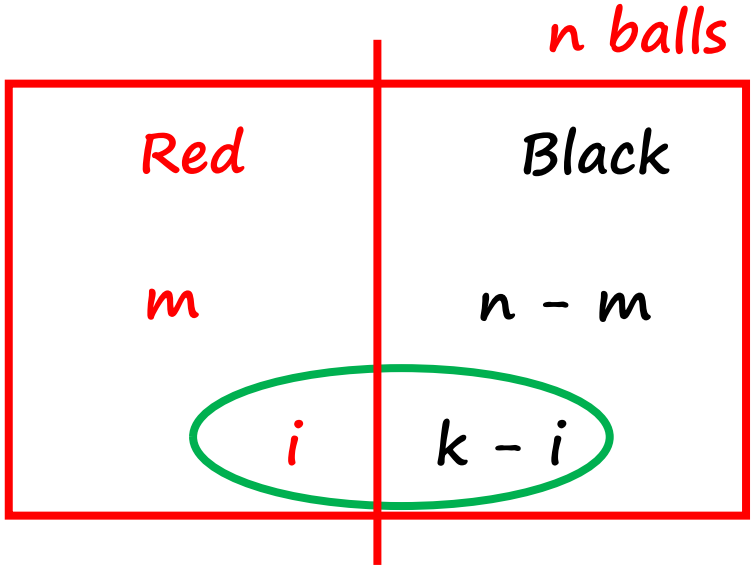
$P(i \text{ red balls}) = P_r$

$|\Omega| = \binom{n}{k}$

➡ Let C = # of ways to get i red balls in k balls

$= a * b$

- a = # of ways to get i out of m red balls
- b = # of ways to get $(k-i)$, out of $(n-m)$, black balls



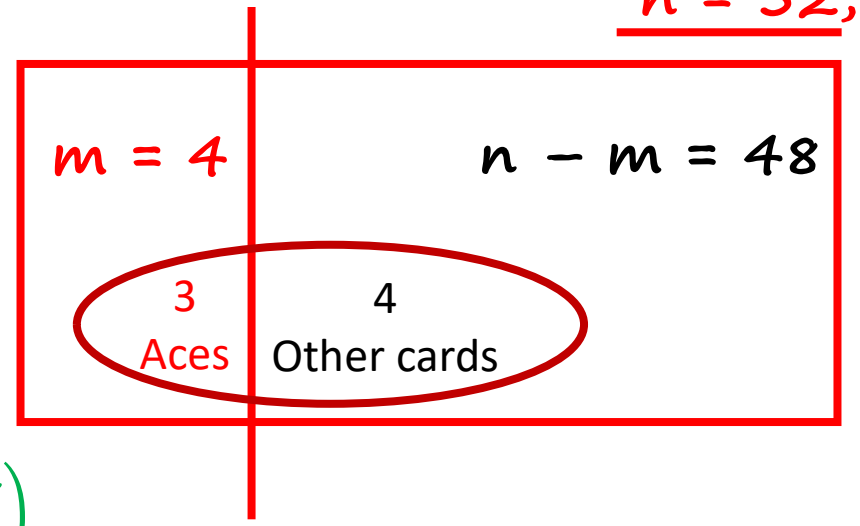
$$P_r = \frac{C}{|\Omega|} = \frac{\binom{m}{i}\binom{n-m}{k-i}}{\binom{n}{k}}$$

Counting Example

In a deck of 52 cards, you select 7 cards. What is the probability that out of 7 cards three are aces, P (3 aces)?

$$P_r = \frac{C}{|\Omega|} = \frac{\binom{m}{i}\binom{n-m}{k-i}}{\binom{n}{k}}$$

$n = 52,\ k = 7,\ i = 3$



$|\Omega| = \binom{n}{k}$

$C = a * b$

$$P\ (3\ aces) = \frac{\binom{4}{3}\binom{48}{4}}{\binom{52}{7}}$$

The multinomial Probabilities

- Balls of different colors: i = 1, . . . , r
- Probability of picking a ball color i is p_i
- Draw n balls, independently
- Given nonnegative numbers n_i, with $n_1 + \ldots + n_r = n$
- Find: $\mathbf{P}(n_1 \; balls \; of \; color \; 1, n_2 \; balls \; of \; colors \; 2, \ldots, n_r balls \; of \; color \; r \;)$
- Special case r = 2; colors: "heads" , "tails"

p 1-p

$$\binom{n}{k} p^k (1-p)^{n-k}$$

n_1 ↔ k heads $p_1 = p$

n_2 ↔ n-k heads $p_2 = 1 - p$

$$\frac{n!}{n_1! \, n_2!} p_1^{n_1} p_2^{n_2}$$

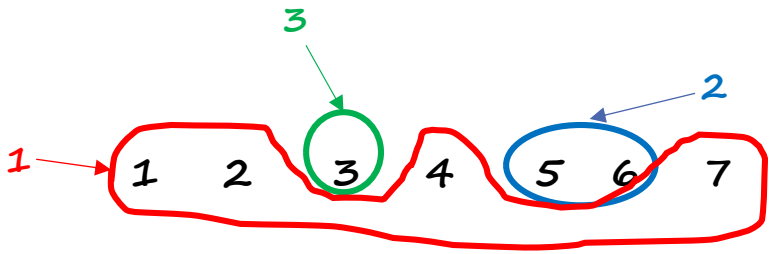
The multinomial Probabilities

$n_1 + \ldots + n_r = n$

r = 3

n = 7

1 1 3 1 2 2 1



Type: (4, 2, 1)

Prob = $p_1 \cdot p_1 \cdot p_3 \cdot p_1 \cdot p_2 \cdot p_2 \cdot p_1 = p_1^4 \cdot p_2^2 \cdot p_3$

$\mathbf{P}$(particular sequence of "type"$(n_1, n_2, \ldots, n_r)$) = $p_1^{n_1} p_2^{n_2} p_3^{n_3} \ldots p_r^{n_r}$

sequence of "type"$(n_1, n_2, \ldots, n_r)$ $\longleftrightarrow$ Partition of {1, . . . , n}} into subsets of sizes $n_1, n_2, \ldots, n_r$

$$\mathbf{P}(\text{ get type } (n_1, n_2, \ldots, n_r)) = \frac{n!}{n_1!\, n_2!\ \ldots n_r!} p_1^{n_1} p_2^{n_2} p_3^{n_3} \ldots p_r^{n_r}$$

Practise Problem 1 –

- An experiment consists of rolling a 20 sided die three times. The number on top of each die is recorded. The numbers are written down in the order in which they are observed. How many possible ordered triples of numbers can result from the experiment? (Note the triple (17, 10, 3) is not the same result as the triple (3, 10, 17)).

Answer:-
There are 20 ways each throw can come up and the order is important so the answer is $20 \cdot 20 \cdot 20 = 20^3 = 8000$.

Practise Problem 2

- When you buy a Powerball ticket, you select 5 different white numbers from among the numbers 1 through 59 (order of selection does not matter), and one red number from among the numbers 1 through 35. How many different Powerball tickets can you buy?

Answer:-

you need to select 5 distinct white numbers, so you can do this $C(59, 5) = 5,006,386$ ways. Then you can pick the red number $C(35, 1) = 35$ ways so the total number of tickets is:-

$C(59, 5) \cdot C(35, 1) = 5,006,386 \cdot 35 = 175,223,510.$

Practise Problem 3

- If I flip a coin 20 times, I get a sequence of Heads (H) and tails (T). How many different sequences have at least three heads?

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Solution: The total number of sequences that are possible here is 2^{20}. We want number of sequences that have at least 3 heads, so we can find out the number of sequences that have no heads, 1 heads, and 2 heads and subtract it from total number of sequences.

$$\text{Number of sequences with no heads} = 1$$

$$\text{Number of sequences with 1 heads} = \binom{20}{1}$$

$$\text{Number of sequences with 2 heads} = \binom{20}{2}$$

Therefore,

$$\text{Number of sequences with at least 3 heads} = 2^{20} - 1 - \binom{20}{1} - \binom{20}{2}$$

We can also approach it by finding the number of sequence with 3 heads, 4 heads, all the way to 20 heads and summing them up, i.e.

$$\text{Number of sequences with at least 3 heads} = \sum_{k=3}^{20} \binom{20}{k}$$

Practise Problem 4

- From 5 seniors, 4 juniors, 2 sophomores and 9 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Answer

There are 5 ways to select a senior, 4 ways to select a junior, 2 ways to select a sophomore and 9 ways to select a freshman. This gives $5 \cdot 4 \cdot 2 \cdot 9 = 360$ ways to select a subset with 4 elements containing one member of each class. When you have done this there are $20 - 4 = 16$ members left and you may choose any one of these to round out the group. Hence the answer is 360 . 16/2 = 2,880. You must divide by 2 because each set of 5 elements selected by this procedure occurs twice

Practise Problem 4 –

Some more explanation

- From 5 seniors, 4 juniors, 2 sophomores and 9 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

Answer

Here is another approach.

Because there are 5 members in the subset and 4 classes, exactly one class occurs twice.

- If there are 2 seniors, these can be selected in $C(5, 2)$ ways and the set filled out with 1 junior, 1 sophomore and 1 freshman, hence in $C(5, 2) \cdot 4 \cdot 2 \cdot 9 =$ 720 ways.
- If there are 2 juniors, these can be selected in $C(4, 2)$ ways and the set filled out with 1 senior, 1 sophomore and 1 freshman, hence in $5 \cdot C(4, 2) \cdot 2 \cdot 9 =$ 540 ways.
- If 2 sophomores, $5 \cdot 4 \cdot C(2, 2) \cdot 9 =$ 180.
- If 2 freshmen, $5 \cdot 4 \cdot 2 \cdot C(9, 2) =$ 1,440.

Hence the answer is $720 + 540 + 180 + 1{,}440 = 2{,}880$.

Practice Problem No 5: Recall that a standard deck of cards has 52 cards. The cards can be classified according to suits or denominations. There are 4 suits, hearts, diamonds, spades and clubs. There are 13 cards in each suit. There are 13 denominations, Aces, Kings, Queens, ... ,Twos, with 4 cards in each denomination. A poker hand consists of a sample of size 5 drawn from the deck. Poker problems are often like urn problems, with a hitch or two.

- Part a: How many poker hands consist of 2 Aces and 3 Kings?
- Part b: If a poker hand has three cards from one denomination and two from another it is called a full house. Find the Probability of getting a Full House, P_{FH}.
- Part c: A royal flush is a hand consisting of an Ace, King, Queen, Jack and Ten, where all cards are from the same suit. Find the Probability of getting a royal flush, P_{RF} .

Practice Problem No 5: Recall that a standard deck of cards has 52 cards. The cards can be classified according to suits or denominations. There are 4 suits, hearts, diamonds, spades and clubs. There are 13 cards in each suit. There are 13 denominations, Aces, Kings, Queens, ... ,Twos, with 4 cards in each denomination. A poker hand consists of a sample of size 5 drawn from the deck. Poker problems are often like urn problems, with a hitch or two.

- **Part a: How many poker hands consist of 2 Aces and 3 Kings?**

Aces $\binom{4}{2} \times$ 3Kings $\binom{4}{3}$

$= 6 \cdot 4 = 24$ ways.

- **Part b: If a poker hand has three cards from one denomination and two from another it is called a full house. Find the Probability of getting a Full House, P_{FH}.**

There are 13 ways to Pick the first denomination and One can choose $\binom{4}{3}$ ways of Selecting 3 cards of that denomination. There are 12 ways of selecting the 2nd denomination and $\binom{4}{2}$ of selecting the 2 cards out of 4.

$$= 13 \cdot \binom{4}{3} \times 12 \times \binom{4}{2} = 3744$$

$$P_{FH} = \frac{3744}{\binom{52}{5}} = \frac{3744 \times 5!}{48 \times 49 \times 50 \times 51 \times 52}$$

$$= 0.00144$$

- **Part c: A royal flush is a hand consisting of an Ace, King, Queen, Jack and Ten, where all cards are from the same suit. Find the Probability of getting a royal flush, P_{RF} .**

There are 4 Royal flush's in the deck and within Each 1 way of Selecth the Cards. (3)

$$\frac{4}{\binom{52}{5}} = \frac{4 \times 120}{48 \times 49 \times 50 \times 51 \times 52}$$

$$= 1.53 \times 10^{-6}$$